Polynomials, Roots, Vieta & Remainders

Student Assimilation Pack | Representation before calculation

RECONNECT → DISCOVER → MAKE SENSE → TRY → DIAGNOSE → FADE → ADOPT → TRANSFER

Polynomials, Roots, Vieta & Remainders

Integrated Assimilation Book

> The requested information chooses the representation.

1. RECONNECT - one polynomial, several useful forms

Consider

P(x)=x^2-5x+6.

Expanded form exposes coefficients:

x^2-5x+6.

Factored form exposes zeros:

(x-2)(x-3).

Root list exposes the same zero information directly:

2,3.

No representation is universally best.

First attempt

For each request, name the cheapest representation before solving.

1. Find the roots of **x^2-5x+6**.
2. Find the sum and product of the roots of **2x^2-5x-3**.
3. Decide how many real roots **x^2-4x+5=0** has.
4. Find the remainder of **x^100** when reduced under **x^2=1**.

2. DISCOVER - roots and factors are the same zero structure

If **P(r)=0**, then **x-r** is a factor of **P(x)**.

For a quadratic with roots **alpha,beta** and leading coefficient **a**,

P(x)=a(x-alpha)(x-beta).

Expanding,

P(x)=a[x^2-(alpha+beta)x+alpha beta].

Compare this with

ax^2+bx+c.

Coefficient matching gives

-a(alpha+beta)=b, so

alpha+beta=-b/a,

and

a alpha beta=c, so

alpha beta=c/a.

This is Vieta derived from factor expansion. It is not an independent formula list.

Use the information the target asks for

For **x^2-7x+10=0**, if the target is **alpha^2+beta^2**, do not solve roots first:

alpha+beta=7, **alpha beta=10**,

so

alpha^2+beta^2=49-20=29.

Contrast

- "find **alpha,beta**" -> individual roots may be needed;
- "find **alpha^2+beta^2**" -> symmetric invariants are enough.

3. MAKE SENSE - discriminant describes quadratic root behavior

For

ax^2+bx+c=0, **a!=0**,

the quadratic formula is

x=(-b +- sqrt(b^2-4ac))/(2a).

The quantity

Delta=b^2-4ac

controls real-root behavior:

- **Delta>0**: two distinct real roots;
- **Delta=0**: one repeated real root;
- **Delta<0**: no real roots.

Root geometry

The parabola **y=ax^2+bx+c** meets the x-axis twice, touches once, or does not meet it according to the same three cases.

Mandatory boundary with the inequality/optimization topic

Question: "How many real roots does **x^2-4x+5=0** have?" Use discriminant: **Delta=16-20=-4**, so none.

Question: "What is the minimum value of **x^2-4x+5**?" That is an optimization request. Complete the square in the inequality/optimization topic: **(x-2)^2+1**, minimum **1**.

Same quadratic, different requested information, different canonical method.

4. DISCOVER - transformed roots and shifted input move in opposite directions

Suppose **P(x)** has roots **alpha,beta**.

To build a polynomial with roots **alpha+c, beta+c**, replace **x** by **x-c**:

Q(x)=P(x-c).

Why? If **x=alpha+c**, then **x-c=alpha**, so **P(alpha)=0**.

Example

P(x)=x^2-5x+6 has roots **2,3**.

Roots shifted up by 4 are **6,7**.

Use

Q(x)=P(x-4)=(x-6)(x-7)=x^2-13x+42.

But **P(x+4)** has roots **-2,-1**.

> Root shift **+c** requires input shift **-c**.

5. MAKE SENSE - remainder theorem and factor theorem

When a polynomial **P(x)** is divided by **x-a**, the remainder is **P(a)**.

Reason: division gives

P(x)=(x-a)Q(x)+r,

where **r** is constant. Set **x=a**:

P(a)=r.

Therefore:

- remainder on division by $x-a$ is $P(a)$;
- $x-a$ is a factor exactly when $P(a)=0$.

Example

For $P(x)=x^3+2x+5$, remainder on division by $x-2$ is

$P(2)=8+4+5=17$.

6. DISCOVER - polynomial reduction is the canonical extension of the prerequisite algebra topic relation rewriting

the prerequisite algebra topic taught the local habit: if an admissible x satisfies $x^2=3x-1$, replace higher powers rather than solving the roots.

this polynomial topic generalizes this to polynomial reduction.

Modulo the relation

$x^2-3x+1=0$,

we may write

$x^2 \equiv 3x-1$.

Every higher power reduces to degree less than **2**.

For example,

$x^3 \equiv x(3x-1) = 3x^2-x \equiv 8x-3$,

and ultimately

$x^5 \equiv 55x-21$.

The language is now about the remainder class modulo a polynomial, not only values at individual roots.

Contrast

To find x^{2026} modulo x^2+x+1, do not calculate 2026 multiplications.

Because

$x^3-1=(x-1)(x^2+x+1)$,

we have

$x^3 \equiv 1 \pmod{x^2+x+1}$.

Since $2026 \equiv 1 \pmod 3$,

$x^{2026} \equiv x$.

7. MAKE SENSE - common-root elimination lowers degree

Suppose a number is a common root of

$x^2-5x+6=0$

and

$x^2-4x+3=0$.

Subtract the equations:

$-x+3=0$.

So any common root must be $x=3$.

Check it in both originals: it works.

This is cheaper than solving two quadratics separately and intersecting the answer sets.

8. TRY - support fades toward independence

Attempt before support.

Full support - execution relation supplied

Roots **alpha,beta** of **x^2-8x+12=0**. Find **alpha^2+beta^2**.

Use Vieta **alpha+beta=8**, **alpha beta=12**, then **alpha^2+beta^2=(alpha+beta)^2-2alpha beta=40**.

Medium support - representation supplied

Find the polynomial whose roots are **5** more than the roots of **x^2-3x-4**.

Cue: use **P(x-5)** and simplify.

Light support - recognition clue

Find the remainder of **x^50** modulo **x^2-1**.

Clue: do not expand powers; reduce the relation first.

Independent - no route supplied

Two monic quadratics share a root: **x^2-7x+10=0**, **x^2-5x+4=0**. Find the common root.

9. DIAGNOSE - four representation errors

Error A - solve roots when target is symmetric

Repair: Vieta first; solve roots only if target still distinguishes them.

Error B - use discriminant for a vertex-value request

Repair: root count -> this polynomial topic discriminant; minimum/maximum -> the inequality/optimization topic square completion.

Error C - shift polynomial input in the same direction as desired root shift

Repair: test one root. Root **alpha+c** must make the input equal **alpha**, hence **P(x-c)**.

Error D - calculate high powers directly

Repair: identify the polynomial relation and reduce after each multiplication, or exploit periodicity.

10. ADOPT - one topic-wide First-Step logic

REQUEST

-> roots individually? -> factor/solve

-> symmetric root information? -> Vieta

-> root count/behavior? -> discriminant

-> transformed roots? -> transform input

-> remainder/factor? -> evaluate or divide

-> high power modulo relation? -> reduce

-> common root? -> eliminate

-> CHECK

11. Historical anchor traces

IOQM-2025-Q16

Combine the given polynomial functions so the root information is visible. The independently verified route reconstructs a quadratic with roots **-2,3**, determines its scale, and recovers the requested value **22**.

IOQM-2025-Q24

Factor the divisor as **(x^2+1)(x^2+x+1)** and reduce **x^2025** modulo the polynomial instead of computing the high power. The independently verified remainder evaluated at the requested point gives **53**.

IOQM-2024-Q24

The coefficient restriction creates a finite structural polynomial problem; exhaustive independent checking of the admissible binary-coefficient configurations confirms answer **50**.

IOQM-2023-Q12

A cubic identity creates a factorized alternative; parity/admissibility eliminates one branch and the remaining equal-structure branch gives answer **18**.

These anchors are mechanism evidence, not weightage evidence.

12. TRANSFER

- symmetric root expressions -> algebraic invariants without explicit roots;
- parameter root-count questions -> discriminant feasibility;
- recurrence/high-power problems -> polynomial reduction / periodic remainders;
- common-root systems -> elimination/GCD thinking;
- transformed-root questions -> input transformation, a useful precursor to functional transformations.

Final belief

> Before solving a polynomial, I ask what information the target needs and choose the representation that exposes exactly that information.

Polynomials, Roots, Vieta & Remainders - First-Step Reference

One topic-wide router

REQUEST

-> individual roots? FACTOR / SOLVE

-> symmetric root target? VIETA

-> root count/real behavior? DISCRIMINANT

-> shifted/transformed roots? TRANSFORM INPUT

-> remainder/factor? EVALUATE / DIVIDE

-> high power modulo relation? REDUCE

-> common root? ELIMINATE

-> CHECK

Recognition atlas

Target clue	First question
alpha+beta, **alpha beta**, symmetric powers	do coefficients already determine it?
number of real roots / repeated root	what is **Delta**?
roots shifted by **c**	should polynomial be **P(x-c)**?
divisor **x-a**	evaluate **P(a)**?
huge exponent + polynomial relation	what low-degree remainder class repeats?
two equations share root	can subtraction/combination lower degree?

Four mandatory contrasts

- solve roots vs use symmetric invariants;
- discriminant/root-count vs vertex/minimum;
- transformed roots vs shifted input;
- calculate high power vs reduce modulo a polynomial.

Polynomials, Roots, Vieta & Remainders - Recognition and First-Line Lab

Write only the first useful line or representation decision.

1. Roots **alpha,beta** of **x^2-7x+10**; target **alpha^2+beta^2**.
2. Find roots of **x^2-7x+10**.
3. Count real roots of **x^2-4x+5=0**.
4. Find minimum value of **x^2-4x+5**.
5. Polynomial **P** has root **3**; what factor follows?
6. Find remainder of **x^3+2x+5** on division by **x-2**.
7. Roots of **P** are **2,3**; desired roots **6,7**. Which input transform?
8. If **Q(x)=P(x+4)**, how do its roots move?
9. Reduce **x^100** modulo **x^2-1**.
10. Reduce **x^2026** modulo **x^2+x+1**.
11. Two quadratics share a root. What operation can lower degree before solving?
12. Roots **r,s** of **2x^2-5x-3**; target **1/r+1/s**. What invariants are needed?
13. Determine whether **x+1** is a factor of **x^3+2x^2-x-2**.
14. A parameter question asks for a repeated real root. What canonical quantity controls it?
15. You need polynomial with roots **2alpha,2beta** where **P(alpha)=P(beta)=0**. What substitution pattern should be tested?
16. Given a huge exponent but no polynomial relation/divisor, can polynomial reduction be invoked yet? State what information is missing.

Polynomials, Roots, Vieta & Remainders - Practice and Transfer Bank

F0 - foundation

1. Factor **x^2-5x+6** and list its roots.
2. For roots **α,β** of **$x^2-7x+10$**, find **$\alpha+\beta$** and **$\alpha\beta$** from factor expansion/Vieta.
3. Find **$\alpha^2+\beta^2$** for the same roots without solving them individually.
4. Compute the discriminant of **x^2-6x+9** and state root behavior.
5. Find the remainder of **x^2+3x+4** on division by **$x-1$**.

F1 - direct

6. For roots of **$2x^2-5x-3$**, find their sum and product.
7. For roots of **$x^2-8x+12$**, find **$\alpha^3+\beta^3$**.
8. Determine root behavior of **$x^2-4x+5=0$**.
9. Determine root behavior of **$x^2-5x+6=0$**.
10. Test whether **$x+1$** is a factor of **x^3+2x^2-x-2**.

F2 - standard

11. **$P(x)=x^2-5x+6$**. Find the monic polynomial whose roots are each **4** greater than the roots of **P**.
12. What are the roots of **$P(x+4)$** if **P** has roots **2,3**?
13. Reduce **x^{100}** modulo **x^2-1**.
14. Reduce **x^{2026}** modulo **x^2+x+1**.
15. Under **$x^2-3x+1=0$**, reduce **x^5** to **$Ax+B$** and express it as a polynomial remainder statement.

F3 - disguised

16. Two equations **$x^2-5x+6=0$** and **$x^2-4x+3=0$** share a root. Find it by elimination.
17. Roots **α,β** of **$3x^2+2x-5$**. Find **$1/\alpha+1/\beta$** without solving roots.
18. Roots **p,q** of **$x^2-sx+t=0$**. Express **p^3+q^3** in **s,t**.
19. Find all **k** for which **$x^2-kx+9=0$** has a repeated real root.
20. For **$P(x)=x^3-4x+3$**, find the remainder upon division by **$x-1$**, and infer whether **$x-1$** is a factor.

F4 - preliminary-style

21. A monic quadratic has roots **α,β**, with **$\alpha+\beta=9$** and **$\alpha^2+\beta^2=41$**. Find **$\alpha\beta$** and the polynomial.
22. If **x^2+x+1** divides a polynomial relation, reduce **$x^{1000}+x^{999}+x^{998}$** modulo **x^2+x+1**.
23. Polynomials **x^2+ax+6** and **x^2+bx+3** share root **-1**. Determine **a,b** and their other roots.
24. Find the remainder of **x^{20}** upon division by **x^2+x+1**.
25. A quadratic has integer coefficients, roots differing by **1**, sum **9**. Reconstruct the roots and monic polynomial using invariant information before expanding.

Transfer

26. Representation change: solve **$\alpha^2+\beta^2$** once by explicit roots and once by Vieta; compare information cost.
27. Boundary contrast: for **x^2-4x+5**, answer both “number of real roots” and “minimum value,” naming the two canonical topics/methods.
28. Transformed-input context: if roots of **P** are unknown but the target roots are **$\alpha-3,\beta-3$**, write the transformed polynomial directly.
29. Recurrence context: explain why a recurrence characteristic relation can turn high powers into low-degree remainders without teaching the recurrence topic here.

30. Common-structure transfer: explain how common-root elimination is a precursor to polynomial gcd thinking.

Polynomials, Roots, Vieta & Remainders - Independent Mixed Mastery Check

No labels or default hints.

1. Roots **alpha,beta** satisfy **2x^2-9x+4=0**. Find **alpha^2+beta^2** without unnecessary root solving.
2. Determine the number of real roots of **3x^2+2x+5=0**.
3. Determine the repeated root of **4x^2-12x+9=0**.
4. **P(x)=x^2-6x+8**. Find the polynomial whose roots are **3** more than the roots of **P**.
5. If **Q(x)=P(x-2)**, describe how every root of **P** moves to a root of **Q**.
6. Find the remainder of **x^4+2x+7** on division by **x-3**.
7. Determine whether **x-2** is a factor of **x^3-3x^2+4**.
8. Reduce **x^99** modulo **x^2+x+1**.
9. Reduce **x^6** modulo **x^2-2x-1**.
10. Two quadratics **x^2-6x+8=0** and **x^2-5x+6=0** share a root. Find it without solving both completely.
11. Roots **r,s** of **x^2-10x+21**. Find **r^3+s^3**.
12. A question asks for the minimum value of **x^2-10x+21**. Route it to the correct canonical topic and state the cheapest representation.
13. A question asks for the values of **k** giving two distinct real roots of **x^2+kx+4=0**. State the governing inequality.
14. WHY NOT: a student expands **x^100** before reducing modulo **x^2-1**. Diagnose the representation error.
15. WHY NOT: a student says roots shifted by **+5** come from **P(x+5)**. Refute by substituting one old root.
16. Transfer: two polynomials share a root and their difference is linear. Explain why solving the linear difference first is logically sufficient only after the candidate is checked in both originals.